

Dear Dr A. Asadi

I am writing to you to apply for the PhD position in Robust Machine Learning for 6G mmWave. I am a 5G engineer working at Accelercomm on the implementation of a power efficient 5G physical layer for use in a satellite direct to device applications.

I am particularly interested in developing my skills in the areas surrounding this PhD, for example in the use of state of the art machine learning developments on increasing both spectrum efficiency while also having the potential to reduce power costs on the ground at base stations. These developments also allow for better quality of service for the end user, which I believe is a challenge worth addressing. At TU Delft, in the WISE lab, there would be a significant step up in the cutting edge communications equipment available to me. But by far the most important aspect for me would be working with fellow researchers and academic staff whom I relish the opportunity to learn from. I believe my personal experience would allow me to fit well in this position. I have been working at Accelercomm for over a year, and I have learnt a great amount in that time. I have been working on improving estimation algorithms and developing a channel processing test framework. I am most proud of this work due to how it has aided my colleagues in helping solve problems and drive development on the channel processing faster than was previously possible. This framework has allowed us to test various spectrum resource allocations in a lower level manner so we can determine accurate performance metrics, for example in a high load scenario using PUSCH with a 64QAM modulation scheme at the maximum number of resource blocks.

At Lancaster University I investigated the work of metal contacts for zinc oxide thin film transistors. This involved creating zinc oxide with spray pyrolysis and depositing metal contacts by physical vapour deposition. I then investigated the electrical characteristics of the transistors. This was my first experience doing research in an academic setting. I found this period very demanding, but this period of my studies the most demanding by far, however I learnt the most during this period about research. I found that even if things were not going right and everything seemed to be going wrong, perseverance was essential in order to reap the benefits of the work. Unfortunately my lectures were not able to get back to me in time before I was able to submit this application. I have attached the contact details for each of them below if you would like further detail about my studies at Lancaster University.

As a PhD researcher at the WISE lab I would want to investigate how the cutting edge research into deep learning and reinforcement learning can be used in the determination of channel state information. I would want to do this in a manner that meets the low power requirements in terrestrial networks while having an improved quality of service. I would investigate the potential use of low power embedded hardware to run reinforcement learning based models for channel estimation in a semi-passive Reconfigurable Intelligent Surface within the 6G mmWave frequency spectrum.

Dr Graeme Burt supervised me during my masters studies at Lancaster working on RF antenna design optimization.

g.burt1@lancaster.ac.uk

+44 (0)1524 592177

Dr Riccardo Degl'Innocenti supervised me during my masters studies at Lancaster with the investigation of THz generation using photoconductive antennas.

r.deglinnocenti@qmul.ac.uk

Dr Jayakrishnan Chandrappan was my manager in the Advanced Packaging department at CSA Catapult during my year long internship with the company.

jayakrishnan.chandrappan@csa.catapult.org.uk

Kinds Regards

Andrew Thomas